

LAB#14

FILE HANDLING IN C++

Lab Tasks

- 1- Write a program in C++ that will:
 - a) Create an empty text file example.txt using ofstream class on default location.
 - b) Assign some values to example.txt file.
 - c) Open example.txt file and print its data on C++ console window.
- 2- Write a program to copy the contents of one file to another.
- 3- Write a program in C++ to count number of lines present in the file example.txt.
- 4- Write a C++ Program to Count Digits, Alphabets and Spaces using File Handling.
- 5- Write a program which outputs its own C++ source file to the screen.

6- What will be the output of the following code?

```
#include <iostream>
#include <fstream>

using namespace std;

int main()
{
    char character;
    int integer;
    ofstream out_stream;
    ifstream in_stream;

    /* Create a file containing two integers */
    out_stream.open("Integers");
    out_stream << 123 << ' ' << 456;
    out_stream.close();

    /* Attempt to read a character, then an integer,
       then a character again, then an integer again,
       from the file "Integers" just created.    */
    in_stream.open("Integers");
    in_stream >> character >> integer;
    cout << "character: " << character << "\n";
    cout << "integer: " << integer << "\n";
    in_stream >> character >> integer;
    cout << "character: " << character << "\n";
    cout << "integer: " << integer << "\n";
    in_stream.close();

    return 0;
}
```

1- Write a program in C++ that will:

- i. Create an empty text file example.txt using ofstream class on default location.
- ii. Assign some values to example.txt file.
- iii. Open example.txt file and print its data on C++ console window.

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main()
{
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    ofstream file;
    file.open("example.txt");
    file << "My name is ali\n";
    file << "2025F-BIT-023";
    file.close();

    ifstream file2;
    file2.open("example.txt");
    string line;

    cout << "data of example.txt:\n\n";
    while (getline(file2, line))
    {
        cout << line << endl;
    }
    file2.close();

    return 0;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023

data of example.txt:

My name is ali
2025F-BIT-023

- 2- Write a program to copy the contents of one file to another.

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main()
{
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    ofstream file;
    file.open("text.txt");
    file << "My name is ali\n";
    file << "2025F-BIT-023";
    file.close();

    ifstream copy;
    copy.open("text.txt");

    ofstream paste;
    paste.open("destination.txt");
    string line;
    while (getline(copy, line))
    {
        paste << line << endl;
    }
    copy.close();
    paste.close();
    cout << "Content Copied...";
    return 0;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Content Copied...

- 3- Write a program in C++ to count number of lines present in the file example.txt.

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main()
{
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    ofstream file;
    file.open("example.txt");
    file << "My name is ali\n";
    file << "2025F-BIT-023";
    file.close();

    ifstream file2;
    file2.open("example.txt");
    string line;
    int count = 0;
    while (getline(file2, line)){
        count++;
    }
    file2.close();

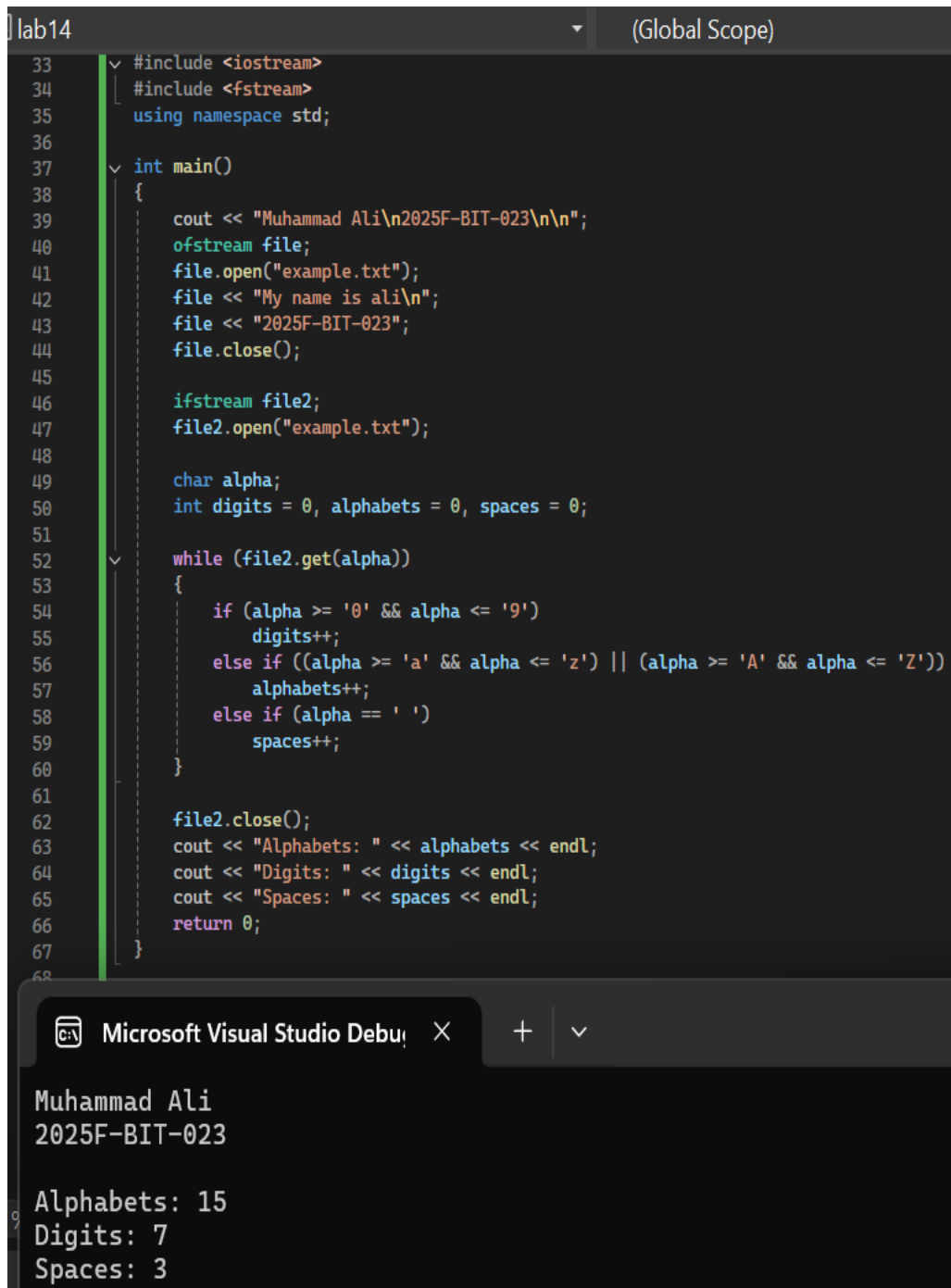
    cout << "Number of lines: " << count;
    return 0;
}
```

Microsoft Visual Studio Debug Console

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2025F-BIT-023

Number of lines: 2

4- Write a C++ Program to Count Digits, Alphabets and Spaces using File Handling.



The screenshot shows a C++ program in Visual Studio. The program opens a file named 'example.txt', reads its contents character by character, and counts the number of digits, alphabets, and spaces. The output is displayed in the console window.

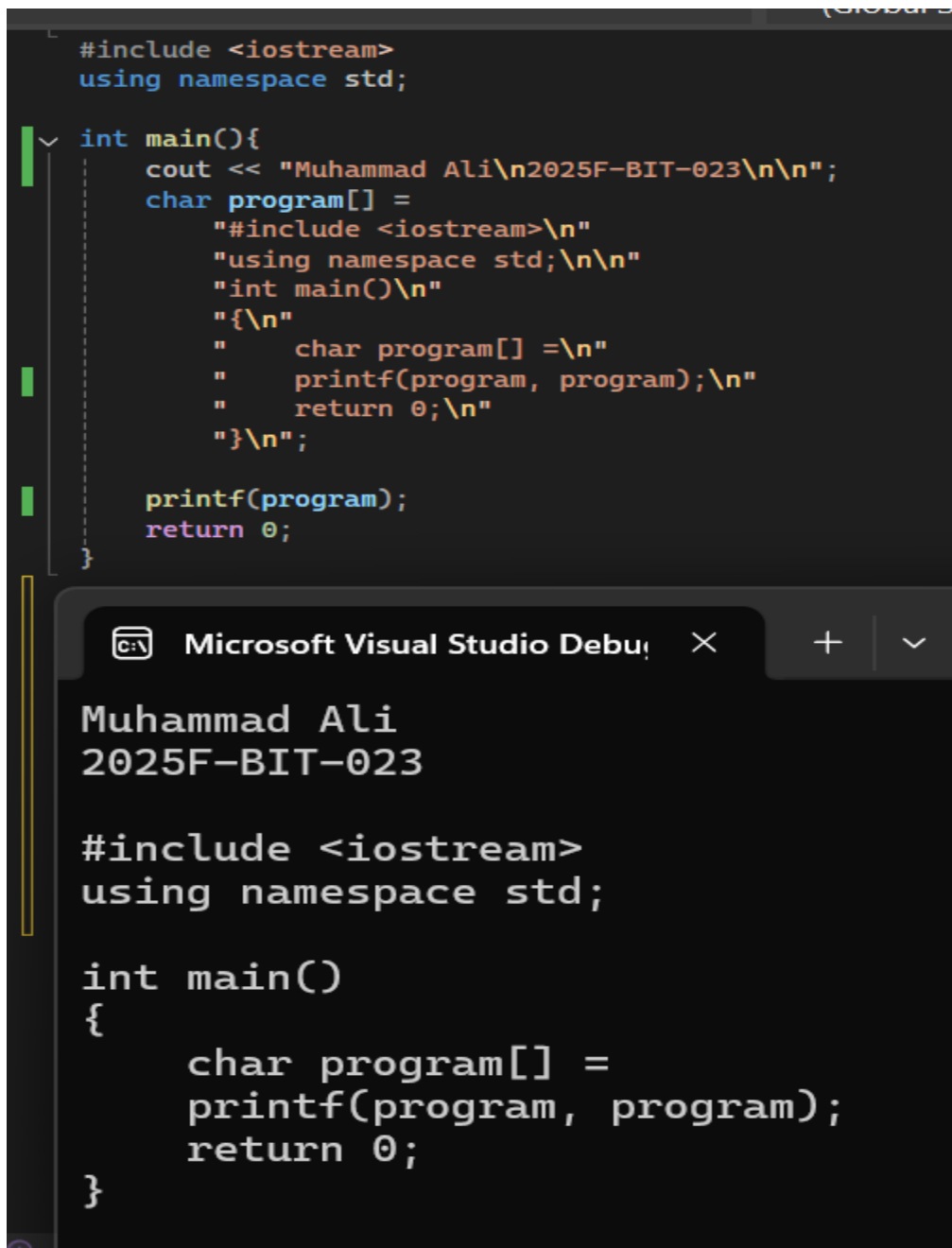
```
lab14 (Global Scope)
33 #include <iostream>
34 #include <fstream>
35 using namespace std;
36
37 int main()
38 {
39     cout << "Muhammad Ali\n2025F-BIT-023\n\n";
40     ofstream file;
41     file.open("example.txt");
42     file << "My name is ali\n";
43     file << "2025F-BIT-023";
44     file.close();
45
46     ifstream file2;
47     file2.open("example.txt");
48
49     char alpha;
50     int digits = 0, alphabets = 0, spaces = 0;
51
52     while (file2.get(alpha))
53     {
54         if (alpha >= '0' && alpha <= '9')
55             digits++;
56         else if ((alpha >= 'a' && alpha <= 'z') || (alpha >= 'A' && alpha <= 'Z'))
57             alphabets++;
58         else if (alpha == ' ')
59             spaces++;
60     }
61
62     file2.close();
63     cout << "Alphabets: " << alphabets << endl;
64     cout << "Digits: " << digits << endl;
65     cout << "Spaces: " << spaces << endl;
66     return 0;
67 }
```

Microsoft Visual Studio Debug Console Output:

```
Muhammad Ali
2025F-BIT-023

Alphabets: 15
Digits: 7
Spaces: 3
```

- 5- Write a program which outputs its own C++ source file to the screen.



The image shows a screenshot of a C++ program in Visual Studio. The program is designed to print its own source code to the console. The code is as follows:

```
#include <iostream>
using namespace std;

int main(){
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    char program[] =
        "#include <iostream>\n"
        "using namespace std;\n\n"
        "int main()\n"
        "{\n"
        "    char program[] =\n"
        "    printf(program, program);\n"
        "    return 0;\n"
        "}\n";

    printf(program);
    return 0;
}
```

The output of the program, shown in the Microsoft Visual Studio Debug Console, is:

```
Muhammad Ali
2025F-BIT-023

#include <iostream>
using namespace std;

int main()
{
    char program[] =
    printf(program, program);
    return 0;
}
```

6- What will be the output of the following code?

```
#include <iostream>
#include <fstream>
using namespace std;
int main()
{
    char character;
    int integer;
    ofstream out_stream;
    ifstream in_stream;
    /* Create a file containing two integers */
    out_stream.open("Integers");
    out_stream << 123 << ' ' << 456;
    out_stream.close();
    /* Attempt to read a character, then an integer,
       then a character again, then an integer again,
       from the file "Integers" just created.    */
    in_stream.open("Integers");
    in_stream >> character >> integer;
    cout << "character: " << character << "\n";
    cout << "integer: " << integer << "\n";
    in_stream >> character >> integer;
    cout << "character: " << character << "\n";
    cout << "integer: " << integer << "\n";
    in_stream.close();
    return 0;
}
```



```
✓ #include <iostream>
  #include <fstream>
  using namespace std;

✓ int main()
  {
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    char character;
    int integer;
    ofstream out_stream;
    ifstream in_stream;
    /* Create a file containing two integers */
    out_stream.open("Integers");
    out_stream << 123 << ' ' << 456;
    out_stream.close();
    /* Attempt to read a character, then an integer, then a character again,
    then an integer again, from the file "Integers" just created.*/
    in_stream.open("Integers");
    in_stream >> character >> integer;
    cout << "character: '" << character << "'\n";
    cout << "integer: " << integer << "\n";
    in_stream >> character >> integer;
    cout << "character: '" << character << "'\n";
    cout << "integer: " << integer << "\n";
    in_stream.close();
    return 0;
  }
```



Microsoft Visual Studio Debug Console



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2025F-BIT-023

character: '1'
integer: 23
character: '4'
integer: 56